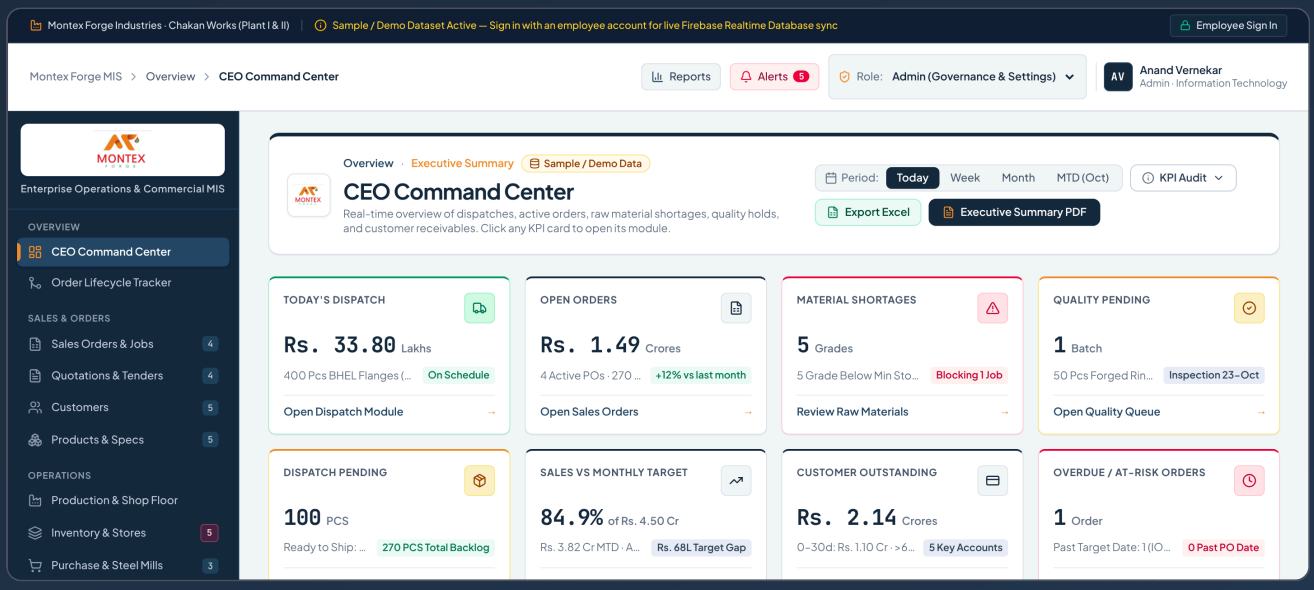
ENTERPRISE MANAGEMENT INFORMATION SYSTEM (MIS)

MONtex Forge Industries — MIS Dashboard

Product Showcase & Digital Transformation Roadmap

One Platform. Connected Operations. Better Decisions.



Live Application Capture: CEO Command Center & Plant Operations Console (1440×900 Desktop Viewport)

Status: Phase 5 Cloud Firestore Architecture Verified

16 Modules

Connected Plant & Commercial Views

Sales, Stores, Purchase, Shop Floor, QC, Dispatch, Finance & Admin

19 Collections

Cloud Firestore Data Model

Includes all 8 persistent operational registers & atomic ledgers

8 RBAC Roles

Zero-Trust Access Governance

Super Admin, CEO, Sales, Stores, Purchase, Quality, Dispatch, Finance

31 / 31 Tests

Automated Verification Suites

Phase 3 Auth, Phase 4 Firestore CRUD & Phase 5 Cutover Verified

Prepared For: **Managing Director, CEO & Management Board**

Application Stack: **React 19, TypeScript, Express Admin API, Cloud Firestore**

Organization: **Montex Forge Industries (Closed-Die & Ring Forgings)**

Document Scope: **Current Capabilities, Honest Status & Stage A/B/C Roadmap**

Presentation Governance Note: This document presents the current application implementation and proposed future enhancements. Screenshots display labeled synthetic preview data to protect confidential customer and plant records.

SECTION 2 — Table of Contents & Feature Classification Governance

Structured navigation guide and transparent implementation status definitions used across this showcase.

MF-MIS-SHOWCASE-2026
Date: October 2026

Complete Document Table of Contents (Clickable Navigation)

19 Executive Sections · 22 Pages

Sec 1. Cover Page & Executive Overview — Page 1

Sec 2. Table of Contents & Classification Key — Page 2

Sec 3. Executive Summary: Purpose & Connected Departments — Page 3

Sec 4. Business Challenges Solved (Before vs. After Matrix) — Page 3

Sec 5. System Overview & Technical Architecture Diagram — Page 4

Sec 6. End-to-End 9-Stage Forging Business Workflow — Page 4

Sec 7.1 Tour: CEO Command Center & Exception Queue — Page 5

Sec 7.2 Tour: 7-Stage Order Lifecycle & Quality/MTC Tabs — Page 6

Sec 7.3 Tour: Sales Orders Register & Customer Master — Page 7

Sec 7.4 Tour: Commercial Quotations & PSU/EPC Tenders — Page 8

Sec 7.5 Tour: Forged Products Catalog & Billet Yard Stock — Page 9

Sec 7.6 Tour: Stock Movement Ledger & Steel Mill POs — Page 10

Sec 7.7 Tour: Approved Steel Mills & Shop Machine Logs — Page 11

Sec 7.8 Tour: Furnace Batch Logs & Maintenance Tickets — Page 12

Sec 7.9 Tour: Plant Power Logs & Quality Lot Inspections — Page 13

Sec 7.10 Tour: Dimensional/TPI Reports & Outward Dispatch — Page 14

Sec 7.11 Tour: Finance, 8-Head Job Costing & Tally Sync — Page 15

Sec 7.12 Tour: Reports Hub, Document Catalog & Alerts — Page 16

Sec 7.13 Tour: Employee RBAC, Audit Trail & Architecture — Page 17

Sec 8 & 9 8-Role Security Matrix & Cloud Firestore Persistence — Page 18

Sec 10 & 11 Business Benefits & Honest Current Limitations — Page 19

Sec 12–14 Future Roadmap: Stages A, B & C Enhancements — Page 20

Sec 15–17 Recommended KPIs, Rollout Steps & Montex Inputs — Page 21

Sec 18 & 19 Executive Glossary & Closing Action Plan — Page 22

Honest Feature Status Classification Standard

Every capability in this document is classified using one of five strict status levels

STATUS BADGE	ENGINEERING & OPERATIONAL MEANING	EXAMPLES IN CURRENT MONTEX MIS BUILD
VERIFIED	Implemented in the codebase and verified passing via automated test suites (Phases 3, 4 & 5 verification scripts).	8-Role RBAC matrix, Admin provisioning API, Firestore CRUD & Cutover Engine, all 8 operational registers, atomic stock/GRN/dispatch writes.
IMPLEMENTED	Present and interactive in the current React UI and wired to context/services, ready for plant UAT.	All 16 workspace views, 6-tab Order Journey details, 8-head Job Costing UI, Tally CSV upload reconciliation, PDF/Excel/CSV client exports.
PARTIALLY IMPLEMENTED	Core screen and workflow exist, but specific statutory/laboratory fields or external connectors remain to be added.	Draft MTC & Tax Invoice PDFs (marked as Draft Summaries pending chemical/mechanical lab fields & e-Invoice IRN/QR fields).
REQUIRES VERIFICATION	Code and architecture are built, but production cutover requires live cloud credentials, company master data, or plant UAT.	Live production Firebase Admin SDK credentials in hosting, cloud deployment of `firestore.rules`, multi-user plant UAT with real Montex data.
PLANNED	Recommended future enhancement proposed in Stage A, B, or C of the Digital Transformation Roadmap.	Scheduled email/WhatsApp digests, background cron alert triggers, direct Tally XML/API 2-way sync, IoT pyrometer/energy meter integration.

Complete Module Inventory & Verification Summary (At a Glance)

- **CEO Command Center:** VERIFIED
- **7-Stage Order Journey:** VERIFIED
- **Sales Orders Register:** VERIFIED
- **Quotations Register:** VERIFIED
- **PSU/EPC Tenders:** VERIFIED
- **Customers & Products:** VERIFIED
- **Billet Yard Inventory:** VERIFIED
- **Stock Movement Ledger:** VERIFIED
- **Steel Mill POs & GRN:** VERIFIED
- **Shift Machine Logs:** VERIFIED
- **Furnace Batch Logs:** VERIFIED
- **Maintenance & Power Logs:** VERIFIED
- **Quality & NDT Lots:** VERIFIED
- **Dimensional & TPI (DIR):** VERIFIED
- **Outward Dispatch & LR:** VERIFIED
- **Receivables & Costing:** VERIFIED
- **Reports & 19-Doc Hub:** VERIFIED
- **8-Role Admin & Audit:** VERIFIED

SECTIONS 3 & 4 — Executive Summary & Business Challenges Solved

Why the Montex Forge MIS was built, how it connects all 8 departments, and the operational friction it eliminates.

MF-MIS-SHOWCASE-2026
Date: October 2026

Section 3A — What the Montex MIS Dashboard Is

- **Unified Industrial Operations Platform:** A purpose-built web application designed specifically for Montex Forge Industries' closed-die and seamless ring-forging operations.
- **Single Source of Truth:** Connects commercial inquiries, customer POs, billet yard stock, steel mill POs, hammer/furnace shifts, quality inspections, outward dispatches, and receivables in one database.
- **Role-Based Departmental Workspaces:** Tailors navigation, write permissions, and document exports across 8 canonical company roles.
- **Real-Time Executive Command Center:** Gives the MD and CEO immediate visibility into daily dispatches, order backlog, critical steel shortages, and overdue customer payments.

Section 3B — Why It Was Built for Montex Forge

- **End-to-End Heat & Job Traceability:** PSU, refinery, and EPC clients (BHEL, IOCL, NTPC, Thermax, L&T) require strict linkage from Steel Mill Heat Number to Forging Batch, MTC, TPI Release, and Invoice.
- **Eliminate Siloed Spreadsheets:** Replaces disconnected departmental Excel sheets and verbal/WhatsApp status updates that cause data mismatches.
- **Protect Order Profitability:** Tracks billet input weights, furnace fuel (PNG/LDO), HT power (kWh/MT), scrap credit, and TPI costs at the individual Job Number level.
- **Zero-Trust Governance:** Enforces active employee authentication, departmental write boundaries, and an immutable security audit log.

Section 3C — Eight Connected Departments on One Shared Platform

1. Top Management (CEO / MD)
Live KPI command center, bottleneck queue, KPI reconciliation, and margin visibility.

2. Sales & Tendering
RFQs, per-kg quotation calculator, PSU tender EMD tracker, and Customer PO registration.

3. Purchase & Procurement
Shortage indents, approved steel mill directory, and Mill Purchase Order tracking.

4. Raw Material Stores
Billet yard balances, job reservations, Mill GRN receipts, and heat-traceable stock movements.

5. Production & Maintenance
Hammer/ring-mill shift logs, furnace temperature/fuel logs, maintenance tickets, and kWh/MT power logs.

6. Quality Assurance & TPI
NDT lot inspections (UT/MPI), dimensional reports (DIR), TPI witnessing, and MTC references.

7. Outward Dispatch
Vehicle gate passes, transporter LR tracking, delivery challans, and automatic PO balance deduction.

8. Finance & IT Admin
0–90+ day receivables ageing, payment receipts, 8-head job costing, Tally CSV sync, and RBAC governance.

Section 4 — Business Challenges Solved: Before vs. After Montex MIS

OPERATIONAL AREA	BEFORE (MANUAL SPREADSHEETS, PAPER DIARIES & WHATSAPP)	AFTER (CONNECTED MONTEX FORGE MIS PLATFORM)
1. Order Status Visibility	Sales must call Stores, Shop Floor, QC, and Dispatch separately to answer a customer's delivery status query.	**7-Stage Order Lifecycle Tracker** shows live PO progress from billet allocation to LR number and invoice on one screen.
2. Quotation & Tender Follow-Up	RFQs and PSU tender deadlines sit in individual emails; repeat-order rates and follow-up dates are easily missed.	**Persistent Quotations & Tenders Registers** auto-calculate per-piece rates from weight, track EMD/bid dates, and convert won quotes to POs.
3. Billet Yard & Heat Control	Physical yard stock is confused with stock already promised to active jobs, causing sudden alloy steel shortages on the press.	**Inventory Engine** separates Physical MT, Reserved MT, and Free Available MT with non-negative stock guards and Mill Heat traceability.
4. Shop-Floor, Furnace & Power Logs	Shift output, furnace soaking temperatures, machine breakdowns, and daily kWh readings remain trapped in paper registers.	**4 Persistent Shop-Floor Registers** log machine shifts, furnace fuel/MT, maintenance root causes, and daily kWh/MT energy intensity.
5. Quality, Dimensional & TPI	NDT lot results, dimensional caliper sheets, and TPI release status are compiled manually right before dispatch.	**Quality & DIR Registers** enforce quantity invariants (`Passed + Rejected <= Inspected`) and link MTC/TPI status directly to the Job No.
6. Dispatch & PO Balance Control	Partial shipments require manual subtraction of shipped pieces from PO quantities, risking over-shipment or billing gaps.	**Atomic Dispatch Transactions** record transporter LR details and automatically deduct shipped pieces from the open Customer PO balance.
7. Receivables & Job Costing	Overdue 60+/90+ day receivables and order-level margin erosion are discovered weeks after month-end closing.	**Live 0–90+ Day Ageing & 8-Head Job Costing** highlight credit-limit breaches immediately and compute net margin per forging job.
8. Access Control & Auditability	Shared spreadsheets allow any user to overwrite formulas, delete rows, or view sensitive financial/pricing data without trace.	**8-Role RBAC & Immutable Audit Trail** restrict write actions by department and log every sensitive change with actor UID and timestamp.

SECTIONS 5 & 6 — System Architecture & End-to-End Business Workflow

How the frontend, role-based security layer, Cloud Firestore database, and reporting engine work together.

MF-MIS-SHOWCASE-2026
Date: October 2026

Section 5 — Multi-Layer Enterprise System Architecture Diagram

VERIFIED

LAYER 1: Role-Aware Enterprise Frontend SPA (React 19 + TypeScript + Tailwind CSS)

16 Interactive Modules

- CEO Command Center
- 7-Stage Order Journey
- Sales Orders
- Quotations & Tenders
- Customers & Products
- Production, Furnace, Maint. & Power
- Inventory & Mill POs
- Quality, DIR & Dispatch
- Finance, Costing, Reports & Admin

LAYER 2: Authentication, 8-Role RBAC & Server-Side Admin SDK Governance

Phase 3 Verified (11/11 Tests)

- Firebase Authentication (ID Tokens)
- 8 Canonical Roles (super_admin to finance)
- Active Account Guard (accountStatus === 'active')
- Express Admin API (/api/admin/employees)
- Self-Lockout & Last-Super-Admin Guards

LAYER 3: Validated Data Service & Controlled Cutover Engine

Phase 4 & 5 Verified (20/20 Tests)

- MontexFirestoreService (Typed CRUD & Pagination)
- MontexCutoverEngine (Dual-Mode Context Bridge)
- Quantity & Stock Invariant Validators
- Duplicate-Submission Locks
- Clean Empty-Collection Isolation (No Mock Leakage)

LAYER 4: Cloud Firestore Named Database (19 Collections) + RTDB Rollback Readiness

DB: ai-studio-montexmis-...

- Core: /users, /customers, /products, /suppliers
- Commercial: /quotations, /tenders, /salesOrders
- Stores: /rawMaterials, /stockMovements, /purchaseOrders
- Shop: /machineLogs, /furnaceLogs, /maintenanceTickets, /powerLogs
- QC/Dispatch/Finance: /qualityInspections, /dimensionalReports, /dispatches, /outstandingAgeing, /jobCostSheets, /auditLogs

LAYER 5: Branded Document & Reporting Engine (jsPDF + SheetJS XLSX + CSV)

19 Classified Document Types

- Official Montex Navy/Orange PDF Header & Logo
- Multi-Sheet Excel Workbooks (.xlsx)
- Standardized CSV Exports
- Readiness Classification Banner (Ready vs. Draft Summary)

Section 6 — End-to-End 9-Stage Montex Forging Business Workflow

Connected data flow from customer inquiry to executive reporting

STAGE 1 · SALES & TENDERING

Inquiry, Quotation & Tender

Log customer RFQs in `/quotations` or PSU bids in `/tenders`; calculate per-piece forging rates from standard weight and follow up to win.

STAGE 2 · ORDER REGISTRATION

Sales Order & Job Number

Convert won quote or register Customer PO in `/salesOrders`; assign internal Job Number (`JOB-MF-2026-XXXX`) and target delivery date.

STAGE 3 · PROCUREMENT

Material Check & Mill PO

Check free available billet stock in `/rawMaterials`; if below reorder level, issue Steel Mill PO in `/purchaseOrders` to approved vendor.

STAGE 4 · STORES & YARD

Mill GRN & Heat Allocation

Receive steel mill delivery via atomic GRN transaction; log Mill Heat Number in `/stockMovements` and allocate billets to the Job Number.

STAGE 5 · SHOP FLOOR

Forging, Furnace & Power

Execute shift production on hammers/ring mills (`/machineLogs`), track furnace soaking & fuel (`/furnaceLogs`), maintenance, and daily kWh/MT.

STAGE 6 · QUALITY & TPI

NDT Lot, Dimensional & MTC

Record UT/MPI lot results in `/qualityInspections`, log drawing measurements in `/dimensionalReports`, and record TPI clearance.

STAGE 7 · OUTWARD LOGISTICS

Dispatch, Gate Pass & LR

Ship cleared pieces in `/dispatches` with transporter & vehicle LR number; atomically deduct shipped quantity from PO balance.

STAGE 8 · FINANCE & COSTING

Invoice, Costing & Ageing

Link GST invoice reference, track 0–90+ day receivables in `/outstandingAgeing`, compute 8-head job margin, and reconcile Tally CSV.

STAGE 9 · EXECUTIVE MIS

CEO Dashboard & Reports

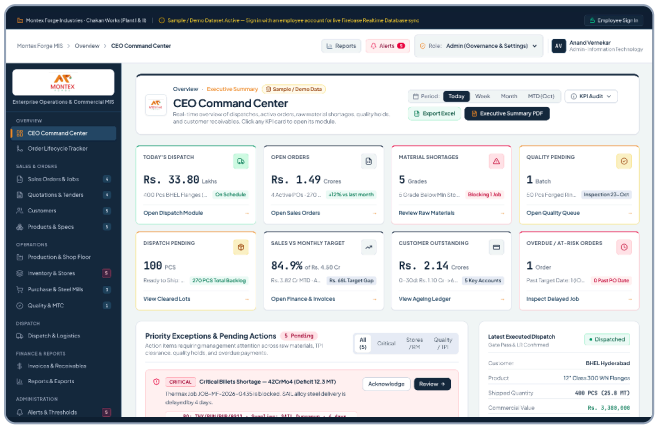
Monitor live plant KPIs, acknowledge threshold alerts in `/actionQueue`, and export branded PDF/Excel management packs.

SECTION 7 — Product Tour: CEO Command Center & Exception Queue

Single-screen executive visibility into daily dispatches, order backlog, material shortages, quality holds, and receivables.

7.1A CEO & Management Command Center — Executive KPIs

VERIFIED



Page Purpose: Give the Managing Director, CEO, and Plant Heads a 30-second health check of commercial and plant performance.

Primary Users: CEO / Managing Director, Super Admin, Department Heads

KEY INFORMATION SHOWN

- Today's Dispatch value (Rs. Lakhs) and piece count against active customer POs.
- Open Orders value (Rs. Crores), active PO count, and undispatched piece balance.
- Critical Raw Material Shortages count and Quality/MTC Pending batch count.
- Sales vs. Monthly Target (%) and Customer Outstanding Receivables (Rs. Crores).

ACTIONS AVAILABLE

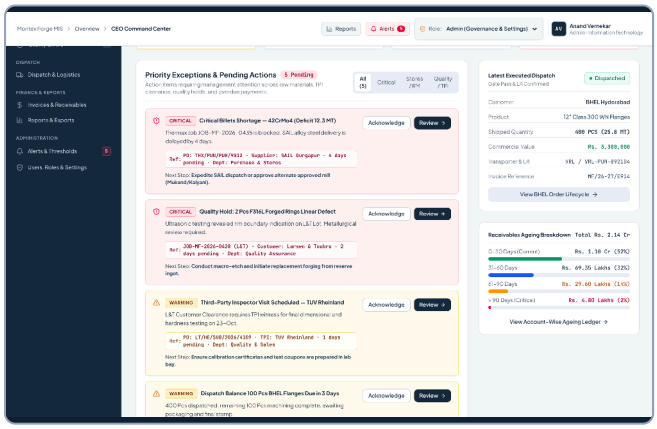
- Switch reporting horizon between Today, Week, Month, and Month-to-Date (MTD).
- Toggle between Computed Live Records and Stored Snapshot with KPI Reconciliation Audit.
- Click any KPI card to drill directly into Dispatch, Sales Orders, Inventory, Quality, or Finance.
- Export the Executive Summary PDF Pack or multi-sheet CEO KPI Excel Workbook (.xlsx).

Real Application Capture (1440×900): Top 8 live KPI cards with Today/Week/Month/MTD period toggles and KPI Reconciliation Audit. *[Synthetic Preview Data]*

Business Value: Eliminates morning status review delays by consolidating 6 departmental spreadsheets into one verified executive view.

7.1B Priority Exceptions, Action Queue & Shop-Floor Pulse

VERIFIED



Page Purpose: Highlight urgent plant bottlenecks (alloy steel shortages, overdue payments, TPI holds) requiring immediate action.

Primary Users: CEO, Plant Head, Stores, Quality, and Finance Heads

KEY INFORMATION SHOWN

- Severity-tagged exception queue (Critical / High / Medium) linked to specific Job Numbers.
- Departmental filter tabs (All, Critical, Stores/RM, Quality/TPI) with pending badge counts.
- Latest Executed Dispatch card showing Customer, Product, Shipped Qty/MT, and Invoice value.
- Shop-floor shift production summary (forged tonnage, furnace utilization, rejection rate).

ACTIONS AVAILABLE

- Acknowledge resolved operational alerts with persistent Cloud Firestore tracking (/actionQueue).
- Click "Review" on any exception to jump directly to the blocked Order Lifecycle Journey.
- Filter action items by responsible department during daily plant coordination meetings.

Real Application Capture (1440×900): Lower section of the CEO Command Center showing severity-ranked plant bottlenecks and latest dispatch confirmation. *[Synthetic Preview Data]*

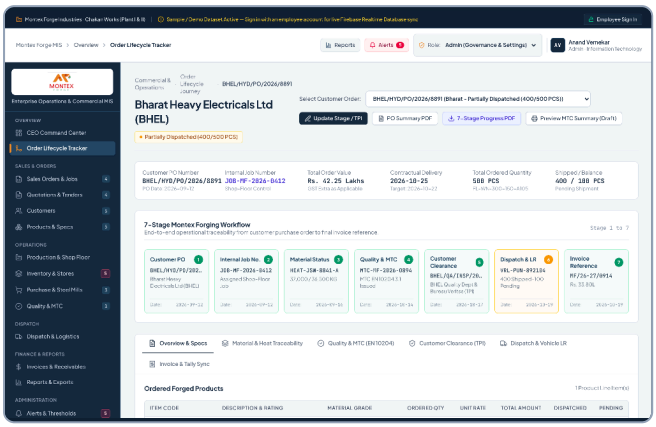
Business Value: Prevents delivery slippages by escalating raw material deficits and TPI holds before hammer/press schedules stall.

SECTION 7 — Product Tour: 7-Stage Order Lifecycle & Job Traceability

End-to-end commercial and technical tracking from Customer Purchase Order to final GST Invoice.

7.2A 7-Stage Order Lifecycle & Internal Job Tracker

VERIFIED



Page Purpose: Track the exact real-time stage of any customer PO across Sales, Stores, Forging, Quality, TPI, Dispatch, and Billing.

Primary Users: Sales, Production Planning, Quality, Dispatch, CEO

KEY INFORMATION SHOWN

- 7-stage progress cards: Customer PO → Internal Job No. → Material Heat → Quality/MTC → TPI Clearance → Dispatch/LR → Invoice.
- Header summary of PO Number, Job Number, Order Value, Contractual Delivery Date, and Shipped vs. Pending Pcs.
- Itemized Ordered Forged Products table with material grade, unit rate, dispatched qty, and balance qty.

ACTIONS AVAILABLE

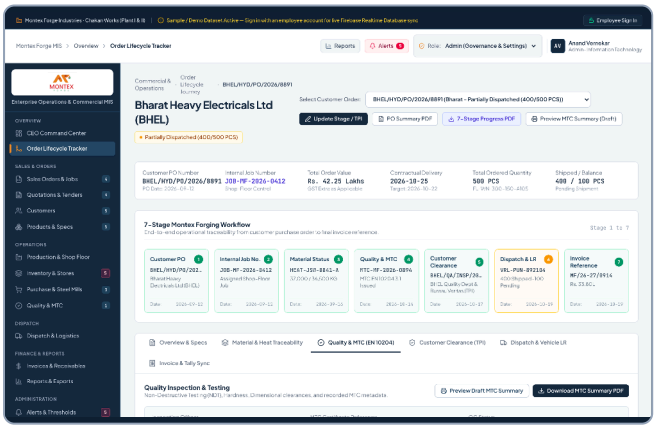
- Select any active Customer Order from the top dropdown to inspect its 7-stage journey.
- Open the "Update Stage / TPI" modal to advance order stage and record TPI clearance reference.
- Generate and preview the PO Summary PDF, 7-Stage Progress PDF, or Draft MTC Summary PDF.

Real Application Capture (1440×900): Interactive 7-stage visual progress pipeline for BHEL/HYD/PO/2026/8891 (Job JOB-MF-2026-0412). *[Synthetic Preview Data]*

Business Value: Allows Sales and Management to answer customer delivery inquiries in seconds without calling the shop floor.

7.2B Order Traceability Sub-Tabs: Material Heat, Quality & TPI

IMPLEMENTED



Page Purpose: Maintain a single audit chain linking steel mill heat numbers, ultrasonic/MPI tests, TPI release notes, and lorry receipts.

Primary Users: Quality Assurance (NDT), TPI Coordinators, Sales, Customer Auditors

KEY INFORMATION SHOWN

- 6 specialized drill-down tabs: Overview & Specs, Material & Heat, Quality & MTC, Customer Clearance (TPI), Dispatch & LR, Invoice.
- Inspection status, MTC reference number, hardness (HBW), ultrasonic testing status, and dimensional check status.
- Third-Party Inspection (TPI) agency details (e.g., Bureau Veritas, Lloyds, EIL) and release certificate date.

ACTIONS AVAILABLE

- Switch across all 6 technical and commercial sub-tabs for the selected Job Number.
- Preview or download the Draft MTC Data Summary PDF with explicit readiness classification.
- Verify whether TPI clearance is approved before authorizing outward gate dispatch.

Real Application Capture (1440×900): Dedicated Quality & MTC (EN 10204) sub-tab inside the Order Lifecycle Tracker linking heat numbers to NDT results. *[Synthetic Preview Data]*

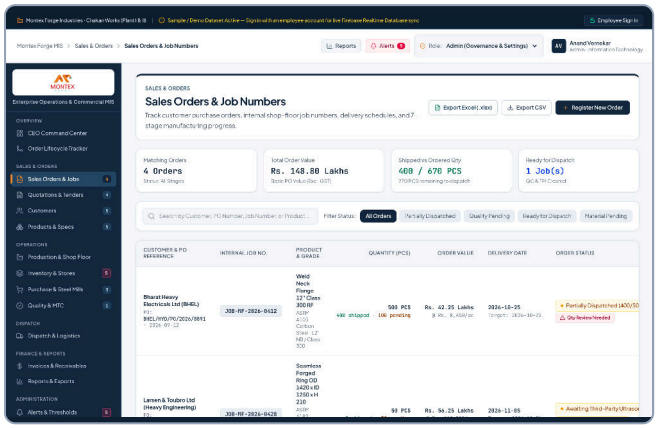
Business Value: Eliminates dispatch of uncleared lots and keeps complete heat-to-invoice traceability ready for PSU/EPC audits.

SECTION 7 — Product Tour: Sales Orders Register & Customer Master

Structured registration of Customer Purchase Orders, internal forging job numbers, and customer credit governance.

7.3A Sales Orders & Internal Job Numbers Register

VERIFIED



Page Purpose: Manage all active, partially dispatched, and completed customer purchase orders with automated quantity validation.

Primary Users: Sales & Marketing, Production Planning, Finance, Management

KEY INFORMATION SHOWN

- Complete register of Order ID, Internal Job No., Customer PO No., Customer, Product, Grade, and PO Value.
- Live piece accounting: Ordered Pcs, Produced Pcs, QC Passed Pcs, Dispatched Pcs, and Pending Balance Pcs.
- Automated Quantity Invariant validation ensuring Dispatched Qty never exceeds QC-Passed or Ordered Qty.

ACTIONS AVAILABLE

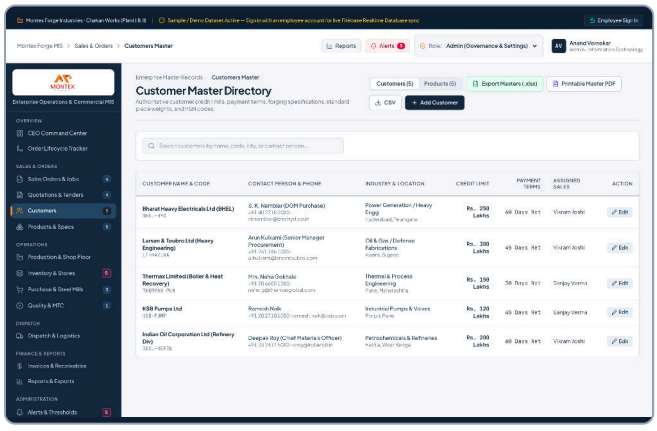
- Search orders by PO Number, Job Number, Customer Name, or Forged Product description.
- Filter by lifecycle stage (PO Registered, Material Allocated, In Production, QC/TPI, Partially Dispatched).
- Register a new Customer Purchase Order via modal form (persisted to Cloud Firestore /salesOrders).
- Export filtered orders to branded Excel (.xlsx), CSV, or Order Progress PDF.

Real Application Capture (1440×900): Central Sales Order register with quantity invariant checks, stage badges, and instant Excel/CSV/PDF exports. *[Synthetic Preview Data]*

Business Value: Prevents over-dispatching, missed delivery dates, and manual numbering errors between Sales and Production.

7.3B Customer Master Directory & Credit Terms

VERIFIED



Page Purpose: Maintain standardized customer records, GST/location details, assigned sales officers, and approved credit terms.

Primary Users: Sales, Finance & Accounts, Super Admin

KEY INFORMATION SHOWN

- Customer Code, Company Name, Primary Plant Location, Industry Sector, and Key Contact details.
- Approved Credit Limit (Rs.) and contractual Payment Terms (e.g., 30 Days, 45 Days, 60 Days).
- Assigned Montex Sales Officer and active account status.

ACTIONS AVAILABLE

- Search customers by company name, code, plant city, or contact person.
- Add new customer profiles or edit credit limits and contact details (persisted to /customers).
- Export the complete Customer & Product Master workbook (.xlsx) or CSV.

Real Application Capture (1440×900): Master directory of PSU, EPC, and OEM customers (BHEL, IOCL, Thermax, L&T, Bharat Forge) with credit limits. *[Synthetic Preview Data]*

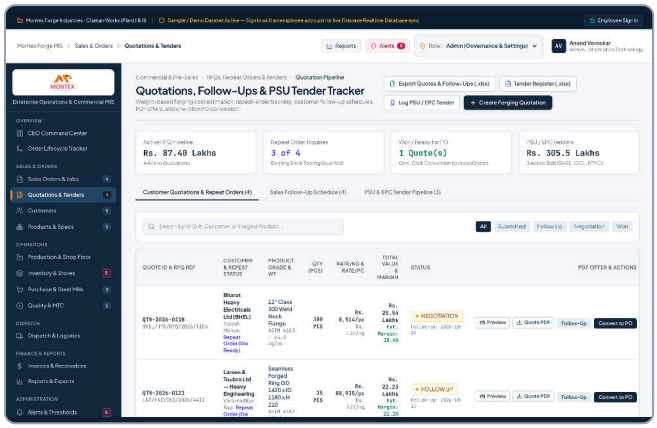
Business Value: Ensures quotations, sales orders, and receivables ageing all reference a single authoritative customer record.

SECTION 7 — Product Tour: Commercial Quotations & PSU/EPC Tenders

Persistent pre-sales pipeline for commercial RFQs, per-kg forging rate calculations, follow-up dates, and PSU tender bids.

7.4A Commercial Quotations & Sales Follow-Up Register (/quotations)

VERIFIED



Page Purpose: Log customer RFQs, compute per-piece forging prices from standard weight and per-kg rates, and track sales follow-ups.

Primary Users: Sales Team, Commercial Estimators, CEO

KEY INFORMATION SHOWN

- Quotation ID, RFQ Reference, Customer, Product, Material Grade, Unit Weight (kg), and Inquiry Qty (Pcs).
- Quoted Rate/kg (Rs.), computed Rate/Pc (Rs.), Total Quoted Value, Estimated Margin (%), and Lead Time (Weeks).
- Repeat-order indicator, Next Follow-Up Date, Follow-Up Notes, and Quote Status (Sent, Follow-Up, Won, Lost).

ACTIONS AVAILABLE

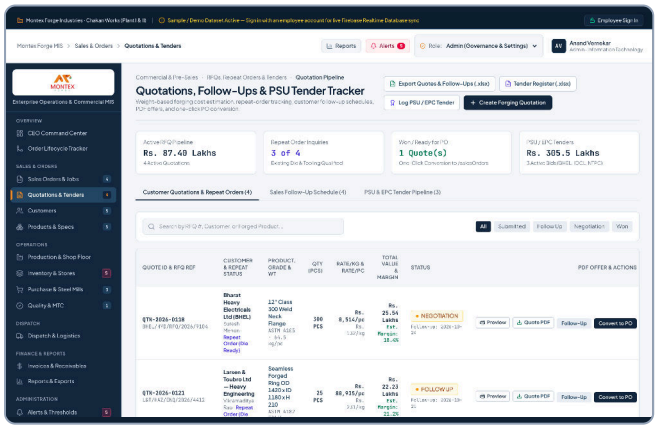
- Create new Commercial Quotations with live per-kg to per-piece rate calculation (saved to /quotations).
- Update follow-up notes and status, or filter solely by pending customer follow-ups.
- Convert a "Won" Quotation directly into an active Sales Order (/salesOrders) with 1 click.
- Preview or download a branded Commercial Quotation PDF or export the register to Excel (.xlsx).

Real Application Capture (1440×900): Persistent Quotations register with automatic weight-to-piece pricing, margin estimation, and 1-click PO conversion. *[Synthetic Preview Data]*

Business Value: Stops RFQ leakage by keeping every quotation, margin calculation, and follow-up date permanently tracked.

7.4B PSU & EPC Tender Bid Pipeline (/tenders)

VERIFIED



Page Purpose: Track high-value government, PSU, and refinery tenders (NTPC, ONGC, IOCL, BHEL, EIL) from bid prep to award.

Primary Users: Tender Cell, Sales Head, Finance (for EMD tracking), CEO

KEY INFORMATION SHOWN

- Tender Number, Issuing Authority, Project Title, Item Summary, and Estimated Contract Value (Rs. Lakhs).
- Earnest Money Deposit (EMD) amount, Bid Submission Deadline, and Price Bid Opening Date.
- Current Tender Stage (Bid Preparation, Submitted, Technical Evaluation, Awarded, Lost).

ACTIONS AVAILABLE

- Register new PSU/EPC tenders into the persistent Cloud Firestore /tenders collection.
- Generate printable Tender Summary Brief PDFs for management bid-review meetings.
- Export the complete Tender Pipeline to Excel (.xlsx) for EMD and bid-calendar tracking.

Real Application Capture (1440×900): Dedicated PSU & EPC Tender register tracking EMD deposits, technical bid dates, and price bid openings. *[Synthetic Preview Data]*

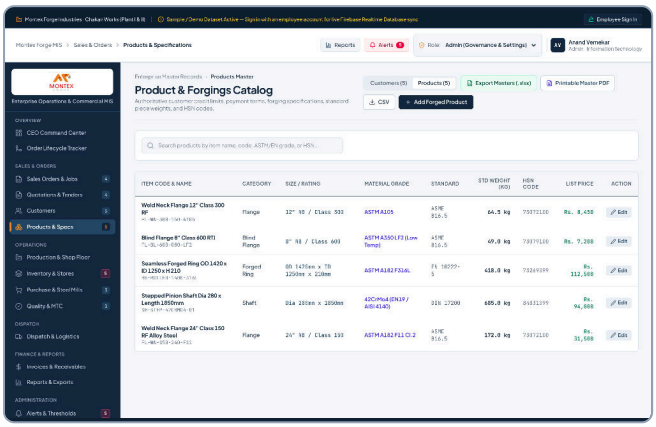
Business Value: Provides clear visibility into upcoming tender submission deadlines and locked EMD capital across PSU bids.

SECTION 7 — Product Tour: Forged Product Specs & Raw Material Yard

Standardized forging catalog weights and real-time billet/ingot yard balances with job reservation guards.

7.5A Forged Products & Technical Specifications Catalog (/products)

VERIFIED



Page Purpose: Define standard finished weights, cut-billet input weights, material grades, dimensional standards, and HSN codes.

Primary Users: Sales, Engineering, Production Planning, Quality

KEY INFORMATION SHOWN

- Product Code, Description, Forging Category (Flanges, Rolled Rings, Shafts, Gear Blanks, Valve Bodies).
- Material Grade (ASTM A105, A182 F11/F22/F316, 42CrMo4, EN24), Dimensional Standard (ASME B16.5 / B16.47).
- Finished Unit Weight (kg), Gross Input Billet Weight (kg), Forging Yield (%), Standard Rate, and HSN Code.

ACTIONS AVAILABLE

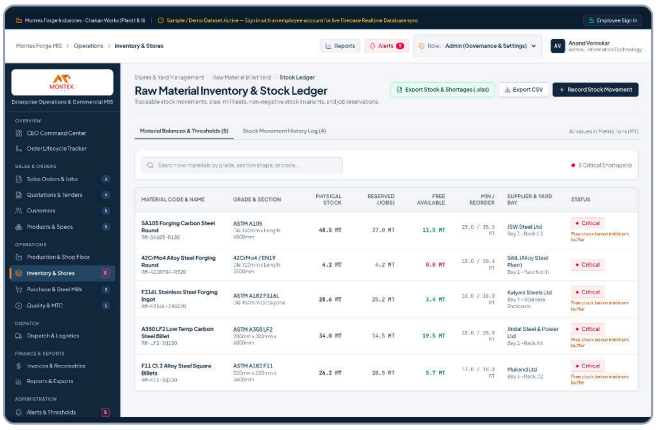
- Search products by item code, flange rating, material grade, or HSN code.
- Add or update forged product specifications in Cloud Firestore (/products).
- Export the Product Technical Catalog to Excel (.xlsx) or CSV.

Real Application Capture (1440×900): Master catalog of closed-die flanges, seamless rolled rings, gear blanks, and valve bodies. *[Synthetic Preview Data]*

Business Value: Standardizes input-to-output weight calculations across quotations, billet requisitions, and job costing.

7.5B Raw Material Billet Yard Balances & Reorder Alerts (/rawMaterials)

VERIFIED



Page Purpose: Monitor alloy and carbon steel billet inventory by grade, section size, and yard bay while preventing stockouts.

Primary Users: Stores Superintendent, Purchase, Production Planning

KEY INFORMATION SHOWN

- Material Code, Grade, Billet Section (mm), Primary Steel Mill Supplier, and Storage Yard Bay.
- Physical Stock (MT), Reserved for Active Jobs (MT), and net Free Available Stock (MT).
- Minimum Safety Stock (MT), Reorder Level (MT), and automatic Critical / Low / Healthy status badges.

ACTIONS AVAILABLE

- Search billet grades or filter to inspect Critical Shortage items blocking active jobs.
- Record new Stock Movements (GRN Receipt or Production Issue) with non-negative stock enforcement.
- Export the 2-sheet Raw Material & Stock Ledger Excel workbook (.xlsx) or CSV.

Real Application Capture (1440×900): Real-time billet and bloom yard stock showing Physical MT, Reserved MT, Free Available MT, and Critical Shortages. *[Synthetic Preview Data]*

Business Value: Separates physical stock from job-reserved stock so Sales and Production never double-allocate the same steel heat.

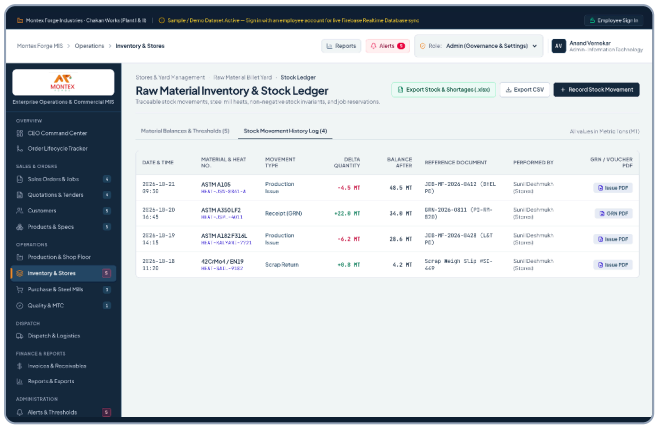
SECTION 7 — Product Tour: Stock Movement Ledger & Steel Mill POs

Atomic inventory transactions and closed-loop steel procurement from Mill Purchase Order to Goods Receipt Note (GRN).

MF-MIS-SHOWCASE-2026
Date: October 2026

7.6A Heat-Traceable Raw Material Stock Movement Ledger (/stockMovements)

VERIFIED



Page Purpose: Maintain a complete, heat-number-traceable audit log of all inward steel receipts and outward forging issues.

Primary Users: Stores, Quality Assurance, Purchase, Internal Audit

KEY INFORMATION SHOWN

- Movement ID, Timestamp, Material Code & Grade, and authoritative Steel Mill Heat Number.
- Movement Type (Receipt GRN vs. Production Issue), Signed Delta Qty (MT), and Resulting Yard Balance (MT).
- Linked Reference Document (GRN No. or Internal Job No.) and operator name.

ACTIONS AVAILABLE

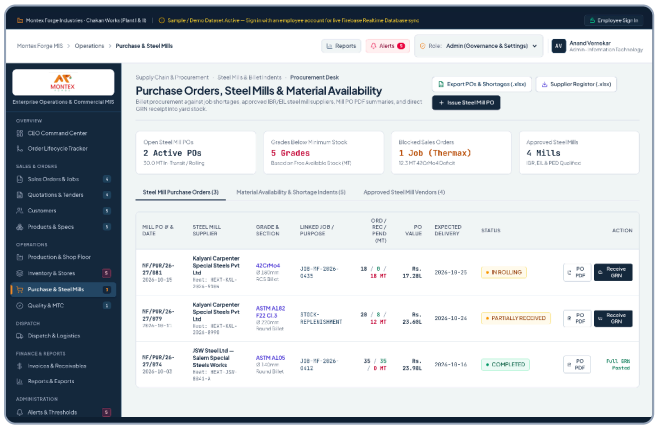
- Log inward GRN receipts or outward production issues via atomic Firestore transaction.
- Generate printable GRN / Material Issue Voucher PDFs directly from any ledger row.
- Export the complete Stock Movement History to CSV or Excel (.xlsx).

Real Application Capture (1440×900): Immutable ledger of every billet receipt (GRN) and shop-floor issue tagged with Steel Mill Heat Numbers. *[Synthetic Preview Data]*

Business Value: Guarantees that yard stock balances and movement history never drift apart, preserving full heat traceability.

7.6B Steel Mill Purchase Orders & GRN Receipt Tracker (/purchaseOrders)

VERIFIED



Page Purpose: Issue purchase orders to approved steel mills and record partial or full GRN receipts directly into yard inventory.

Primary Users: Purchase Department, Stores (GRN Receipt), Finance, CEO

KEY INFORMATION SHOWN

- Mill PO Number, PO Date, Steel Mill Supplier, Material Grade, Section (mm), and Linked Job Number.
- Ordered Tonnage (MT), Received Tonnage (MT), Pending Balance (MT), Rate/MT, and Total PO Value (Rs.).
- Expected Delivery Date, Mill Heat Number, and PO Status (Issued, Partially Received, Completed).

ACTIONS AVAILABLE

- Issue new Steel Mill Purchase Orders persisted to Cloud Firestore (/purchaseOrders).
- Click "Receive GRN" on any open PO to atomically update PO received tonnage, yard stock, and stock ledger.
- Switch to the Shortage Indents tab to see auto-calculated reorder quantities for critical grades.
- Preview/download branded Steel Mill PO PDFs or export the 3-sheet Procurement Excel Workbook.

Real Application Capture (1440×900): Persistent Steel Mill PO register tracking ordered tonnage, received GRN tonnage, pending balance, and heat numbers. *[Synthetic Preview Data]*

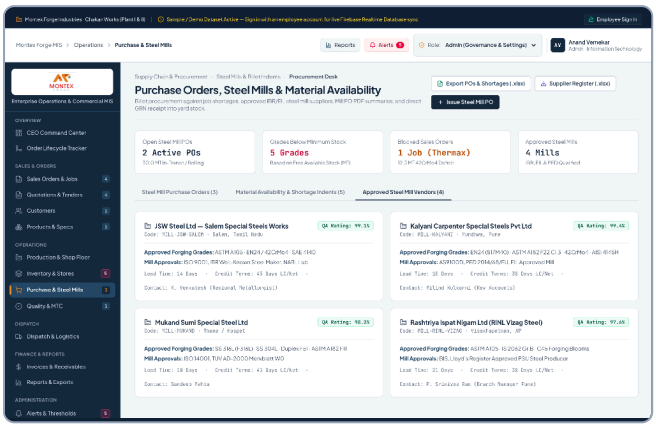
Business Value: Eliminates manual re-entry between Purchase and Stores by linking Mill PO receipts directly to yard inventory.

SECTION 7 — Product Tour: Approved Steel Suppliers & Shop Machine Logs

Qualified steel mill directory and shift-wise hammer, press, and ring-rolling production tracking.

7.7A Approved Steel Mill & Vendor Directory (/suppliers)

IMPLEMENTED



Page Purpose: Maintain a qualified vendor list with approved alloy/carbon steel grades, lead times, credit terms, and quality ratings.

Primary Users: Purchase, Quality Assurance, Management

KEY INFORMATION SHOWN

- Supplier Code, Mill Name, Plant Location, and Primary Contact details.
- Approved Steel Grades (ASTM A105, A182 F11/F22, 42CrMo4, SS 304/316L) and Statutory Certifications (IBR, PED, NABL).
- Standard Delivery Lead Time (Days), Credit Terms (Days), and Historical Quality Rating (%).

ACTIONS AVAILABLE

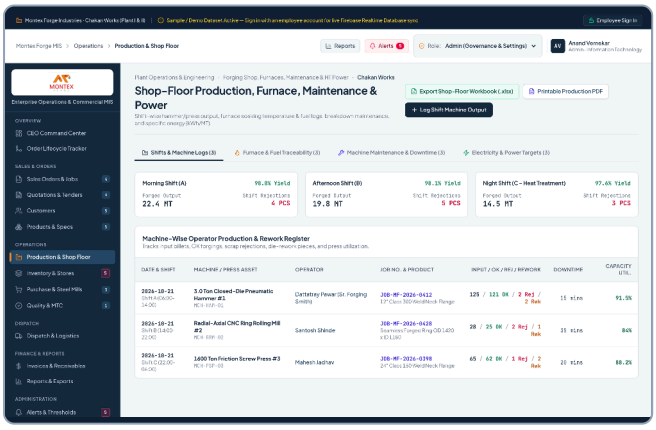
- Inspect approved grades and statutory certifications before placing a steel mill PO.
- Export the Approved Steel Mill Supplier Directory to Excel (.xlsx).

Real Application Capture (1440×900): Directory of IBR, EIL, and PED certified primary steel mills (JSW, Kalyani Carpenter, SAIL, Tata Steel, Mukand). *[Synthetic Preview Data]*

Business Value: Prevents procurement of uncertified raw material heats for regulated PSU and refinery orders.

7.7B Shop-Floor Shift & Hammer/Press Production Logs (/machineLogs)

VERIFIED



Page Purpose: Capture shift-wise billet input, forged output, OK pieces, scrap rejections, rework, downtime, and press utilization.

Primary Users: Shop-Floor Supervisors, Production Manager, Plant Head

KEY INFORMATION SHOWN

- Shift summary cards (Shift A, Shift B, Shift C) showing total forged tonnage (MT) and shift yield (%).
- Machine-wise log: Date, Shift, Asset ID/Name, Forging Operator, Linked Job No., and Product Name.
- Input Billets (Pcs), Forged Output (Pcs), OK Pcs, Rejected Pcs, Rework Pcs, Downtime (Min), and Utilization (%).

ACTIONS AVAILABLE

- Log new Shift Machine Output entries via validated modal form (persisted to /machineLogs).
- Export the 4-sheet Shop-Floor Operations Excel Workbook (.xlsx) or Printable Production PDF.

Real Application Capture (1440×900): Persistent shift register for pneumatic hammers, friction screw presses, and CNC ring-rolling mills. *[Synthetic Preview Data]*

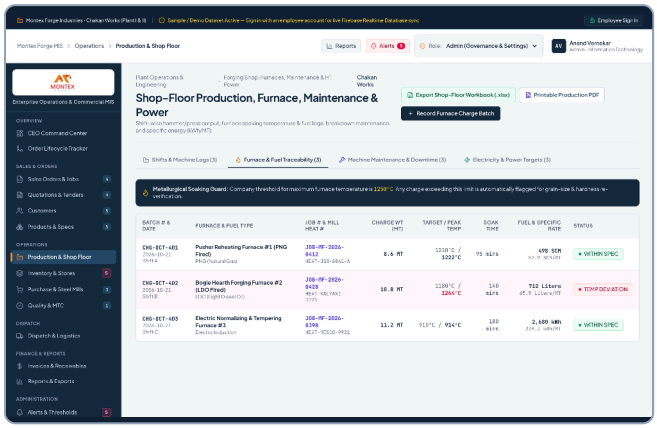
Business Value: Replaces paper shift diaries with structured digital records, exposing press bottlenecks and shift rejection spikes.

SECTION 7 — Product Tour: Furnace Batch Logs & Maintenance Tickets

Persistent thermal traceability for reheating/heat-treatment furnaces and breakdown/preventive maintenance tracking.

7.8A Reheating & Heat-Treatment Furnace Batch Logs (/furnaceLogs)

VERIFIED



Page Purpose: Record thermal cycle parameters and specific fuel consumption per MT for every forging and normalizing/quenching batch.

Primary Users: Furnace Operators, Metallurgists, Quality Assurance, Plant Head

KEY INFORMATION SHOWN

- Batch Number, Date, Shift, Furnace Asset Name, Fuel Type (PNG Natural Gas / LDO / Electric), and Linked Job & Heat No.
- Charge Weight (MT), Target Temp (°C), Peak Recorded Temp (°C), and Soaking Duration (Minutes).
- Total Fuel Consumed (SCM / Liters / kWh), Specific Fuel per MT, and automatic Over-Temp Deviation flags.

ACTIONS AVAILABLE

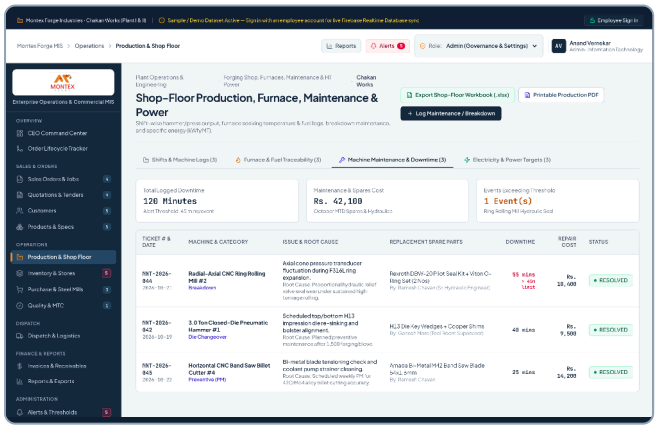
- Log new Furnace Charge & Soaking Batches into Cloud Firestore (/furnaceLogs).
- Automatically flag batches exceeding the company furnace temperature threshold (e.g., >1240°C).
- Include all furnace thermal batches in the multi-sheet Shop-Floor Excel Workbook export.

Real Application Capture (1440×900): Persistent furnace register tracking charge weight, target vs. peak temperature (°C), soaking time, and PNG/LDO fuel use. *[Synthetic Preview Data]*

Business Value: Controls fuel cost per ton of steel forged and provides metallurgical soaking proof for EN 10204 3.1 certificates.

7.8B Machine Breakdown & Preventive Maintenance Tickets (/maintenanceTickets)

VERIFIED



Page Purpose: Track unplanned equipment breakdowns and preventive maintenance jobs across hammers, ring mills, saws, and furnaces.

Primary Users: Maintenance Engineers, Shop-Floor Supervisors, Plant Head

KEY INFORMATION SHOWN

- Ticket ID, Date, Machine ID & Asset Name, and Maintenance Type (Breakdown, Preventive, Die Changeover).
- Issue Description, Engineering Root Cause, and Replacement Spare Parts used.
- Downtime Duration (Minutes), Maintenance Cost (Rs.), Assigned Technician, and Ticket Status (Open, In Progress, Resolved).

ACTIONS AVAILABLE

- Log new Maintenance & Downtime Tickets into Cloud Firestore (/maintenanceTickets).
- Identify recurring mechanical/hydraulic failure causes and high-downtime forging assets.

Real Application Capture (1440×900): Persistent maintenance register logging hammer/press breakdowns, root causes, spares replaced, downtime, and repair cost. *[Synthetic Preview Data]*

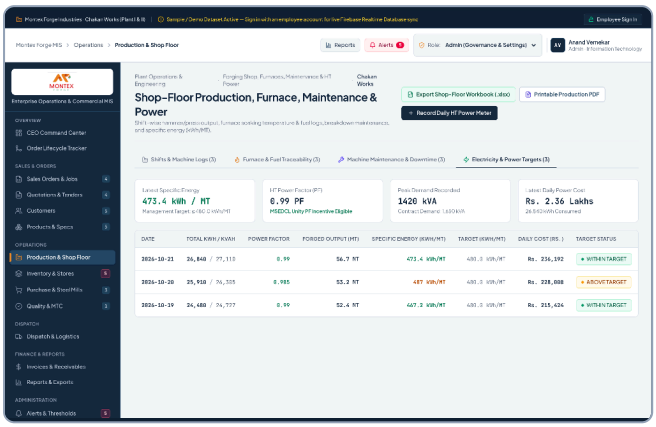
Business Value: Reduces unplanned hammer and ring-mill downtime by creating a permanent maintenance and spares history per asset.

SECTION 7 — Product Tour: Plant Power Logs & Quality Lot Inspections

Specific electricity consumption (kWh/MT) monitoring and non-destructive testing (NDT) lot clearance.

7.9A Electricity & Specific Power Consumption Register (/powerLogs)

VERIFIED



Page Purpose: Monitor daily plant electricity consumption against forged output tonnage to control energy cost per MT and maintain unity power factor.

Primary Users: Electrical Maintenance, Plant Head, Finance/Costing, CEO

KEY INFORMATION SHOWN

- Meter Log Date, Daily Total kWh, Apparent Energy kVAh, and computed Power Factor (PF).
- Peak Demand Recorded (kVA vs. Sanctioned Load), Daily Forged Output (MT), and Specific Power (kWh/MT).
- Daily Energy Cost (Rs.) and automatic alert badge when kWh/MT exceeds threshold (e.g., >480 kWh/MT).

ACTIONS AVAILABLE

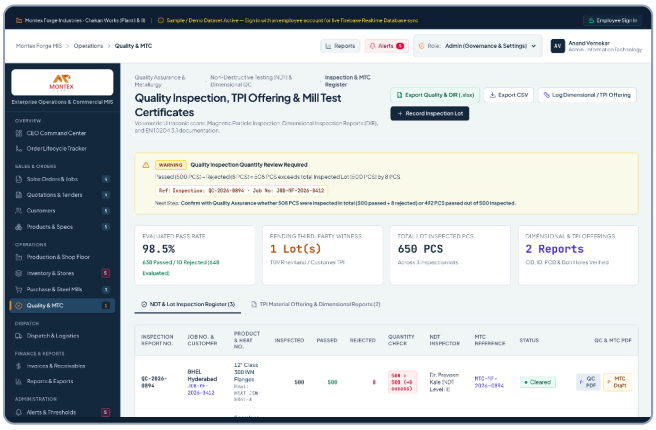
- Record daily HT meter readings into Cloud Firestore (/powerLogs) with auto-computed PF and kWh/MT.
- Export daily power and specific energy logs to Excel (.xlsx) for monthly MSEB/utility bill reconciliation.

Real Application Capture (1440×900): Daily HT electricity register tracking kWh, kVAh, Power Factor, Peak Demand (kVA), and Specific Energy (kWh/MT). *[Synthetic Preview Data]*

Business Value: Prevents power-factor penalties and highlights idle furnace/compressor energy waste on low-tonnage shifts.

7.9B Quality Lot Inspections, NDT & MTC Register (/qualityInspections)

VERIFIED



Page Purpose: Record lot-wise Ultrasonic (UT), Magnetic Particle (MPI), Hardness, and visual inspection outcomes and issue MTC references.

Primary Users: Quality Assurance Engineers (NDT Level II/III), Sales, Dispatch

KEY INFORMATION SHOWN

- Top KPI strip: Total Inspected Pcs, Passed Pcs, Rejected Pcs, and overall Plant First-Pass Yield (%).
- Inspection ID, Date, Job Number, Customer, Product, Grade, and linked Steel Mill Heat Number.
- Inspected Qty, Passed Qty, Rejected Qty, Quantity Invariant Status, NDT Inspector, MTC Reference, and Lot Status.

ACTIONS AVAILABLE

- Log new Quality Lot Inspections into Cloud Firestore (/qualityInspections) with strict quantity validation.
- Preview/download Quality Inspection Lot PDFs and Draft EN 10204 3.1 MTC Summary PDFs.
- Export the 2-sheet Quality & Dimensional Excel Workbook (.xlsx) or CSV.

Real Application Capture (1440×900): Quality Assurance register with enforced quantity invariants (Passed + Rejected <= Inspected) and MTC references. *[Synthetic Preview Data]*

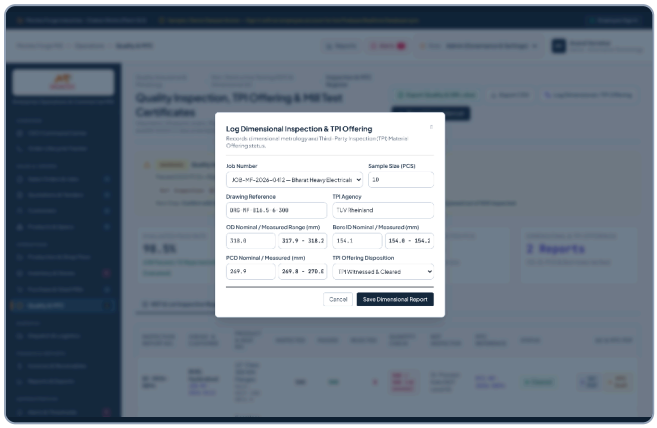
Business Value: Enforces a hard quality gate so only inspected, passed, and MTC-referenced lots advance to outward dispatch.

SECTION 7 — Product Tour: Dimensional/TPI Reports & Outward Dispatch

Drawing-level dimensional verification, Third-Party Inspection (TPI) witnessing, and outward vehicle/LR gate control.

7.10A Dimensional Inspection Reports (DIR) & TPI Witnessing (/dimensionalReports)

VERIFIED



Page Purpose: Document drawing-specific dimensional measurements (OD, ID/Bore, Thickness, PCD, Bolt Holes) and TPI agency sign-off.

Primary Users: QC Inspectors, TPI Engineers (Lloyds, Bureau Veritas, TUV, EIL), Sales

KEY INFORMATION SHOWN

- DIR ID, Inspection Date, Linked Job No., Customer, Product, Engineering Drawing No., and Sample Size.
- Parameter tolerance table: Characteristic (OD, Bore, PCD), Nominal mm, Tolerance, Measured Range, and OK/Out-of-Spec badge.
- Assigned TPI Agency and Offering Status (Internal Cleared, Offered to TPI, Witnessed & Cleared, Hold).

ACTIONS AVAILABLE

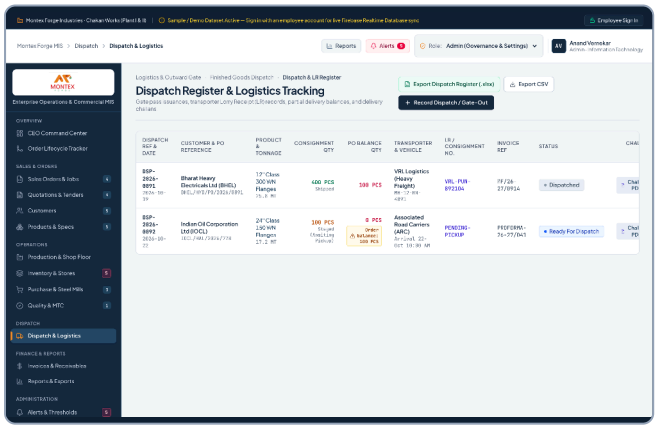
- Create new Dimensional Inspection Reports persisted to Cloud Firestore (/dimensionalReports).
- Preview and download formatted Dimensional Inspection Report (DIR) PDFs for customer submission.

Real Application Capture (1440×900): Persistent Dimensional Inspection Report register storing nominal vs. measured OD, Bore, PCD, and TPI witness status. *[Synthetic Preview Data]*

Business Value: Digitizes caliper/CMM inspection sheets so TPI release documentation is permanently linked to the job number.

7.10B Outward Dispatch, Gate Pass & Lorry Receipt Register (/dispatches)

VERIFIED



Page Purpose: Record outward factory dispatches, update remaining customer PO balances atomically, and generate Delivery Challans.

Primary Users: Dispatch & Logistics Officer, Security Gate, Sales, Finance (Billing)

KEY INFORMATION SHOWN

- Dispatch ID, Date, Customer PO No., Internal Job No., Customer Name, and Forged Product.
- Shipped Consignment Qty (Pcs), Remaining PO Balance (Pcs), and Consignment Weight (MT).
- Transporter Name, Vehicle Registration No., Lorry Receipt (LR) No., Invoice Ref, and Transit Status.

ACTIONS AVAILABLE

- Record new Outward Dispatches (atomically updating /dispatches and the parent /salesOrders balance).
- Preview and download branded Outward Delivery Challan / Gate Pass PDFs.
- Export the complete Dispatch & LR Register to Excel (.xlsx) or CSV.

Real Application Capture (1440×900): Outward logistics register tracking shipped quantities, PO balances, consignment tonnage, transporters, and LR numbers. *[Synthetic Preview Data]*

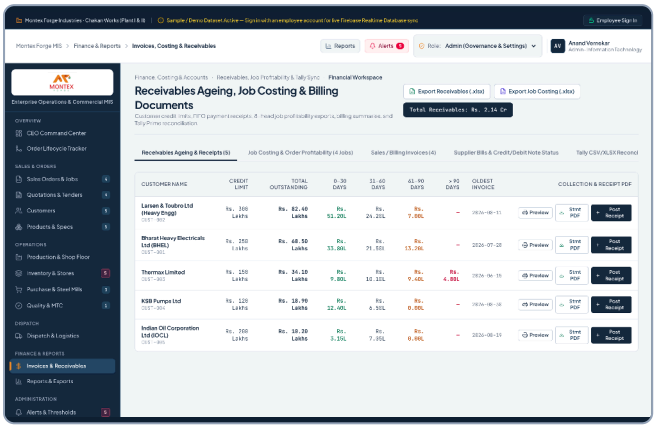
Business Value: Prevents over-shipment beyond ordered quantities and gives Finance immediate proof of dispatch for GST invoicing.

SECTION 7 — Product Tour: Finance, Job Costing & Tally Reconciliation

Receivables ageing buckets (0–90+ days), 8-head order profitability costing, and Tally Prime CSV ledger reconciliation.

7.11A Receivables 0–90+ Day Ageing & Payment Collection (/outstandingAgeing)

VERIFIED



Page Purpose: Monitor customer payment exposure across ageing buckets, enforce credit limits, and record incoming RTGS/NEFT collections.

Primary Users: Finance & Accounts, Sales Head, CEO / Managing Director

KEY INFORMATION SHOWN

- Customer Name, Approved Credit Limit (Rs.), Total Outstanding (Rs.), and Credit-Limit Exceeded warnings.
- Bucketed breakdown: 0–30 Days (Current), 31–60 Days, 61–90 Days, and 90+ Days (Critical Overdue).
- Session Payment Receipts register with UTR reference numbers and downloadable Payment Receipt Vouchers.

ACTIONS AVAILABLE

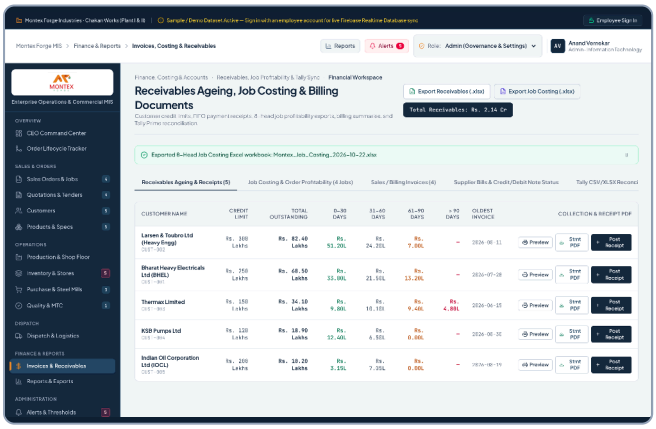
- Click "Record Payment" on any customer row to apply incoming collections (FIFO across oldest ageing buckets).
- Preview/download official Payment Receipt Voucher PDFs, Tax Invoice Draft PDFs, and Supplier PO Summaries.
- Export the complete Receivables Ageing Schedule to Excel (.xlsx).

Real Application Capture (1440×900): Customer-wise receivables schedule with 0–30, 31–60, 61–90, and 90+ day critical buckets and credit-limit alerts. *[Synthetic Preview Data]*

Business Value: Accelerates working-capital recovery by giving Finance and Sales a shared, real-time view of overdue 60+ and 90+ day accounts.

7.11B 8-Head Order Job Costing & Tally CSV Reconciliation (/jobCostSheets)

IMPLEMENTED



Page Purpose: Compare actual per-job manufacturing costs against invoiced revenue and reconcile MIS billing with Tally Prime exports.

Primary Users: Cost Accountant, Finance Head, CEO

KEY INFORMATION SHOWN

- 8 cost heads per Job: Raw Material, Scrap Credit (-), Forging Labour, Electricity, Furnace Fuel, Machining, Testing/TPI, Freight & Overheads.
- Computed Total Job Cost (Rs.), Cost per Kg, Cost per Piece, Gross Margin (Rs.), and Net Margin (%).
- Dedicated Tally CSV/XLSX Reconciliation tab (Screenshot 24) to match MIS invoices against Tally voucher exports.

ACTIONS AVAILABLE

- Select any Job Number to inspect its 8-head cost breakdown or download a Job Cost Sheet PDF.
- Upload/reconcile Tally Prime sales ledger CSV exports in the Tally Reconciliation tab.
- Export the All-Orders Costing & Margin Summary to Excel (.xlsx).

Real Application Capture (1440×900): Granular 8-head job costing sheet computing net margin per order alongside Tally Prime CSV import reconciliation. *[Synthetic Preview Data]*

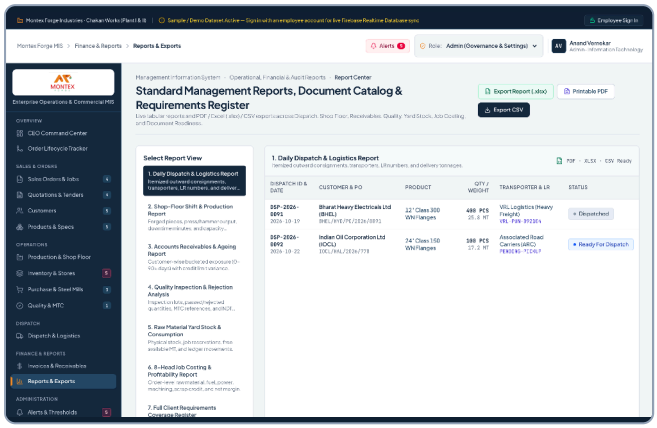
Business Value: Reveals true per-order profitability after accounting for die setup, furnace fuel, power, scrap recovery, and TPI fees.

SECTION 7 — Product Tour: Reports Hub, Document Catalog & Alerts

Centralized PDF/Excel/CSV reporting center, 19-document readiness classification, and configurable plant alert rules.

7.12A Enterprise Reports & 19-Document Readiness Hub (/reports)

VERIFIED



Page Purpose: Provide one-click generation of standardized management reports and transparent readiness classification for all 19 plant documents.

Primary Users: All Authorized Roles (Role-Checked Export Permissions), CEO, Auditors

KEY INFORMATION SHOWN

- 8 interactive report views: Daily Dispatch, Shop-Floor Production, Receivables Ageing, Quality/Rejection, Stock Ledger, Job Costing, Requirements Register, and Document Readiness Catalog.
- Authoritative classification of all 19 MIS documents into "Ready Now" vs. "Draft Summary — Missing Statutory/Lab Fields".

ACTIONS AVAILABLE

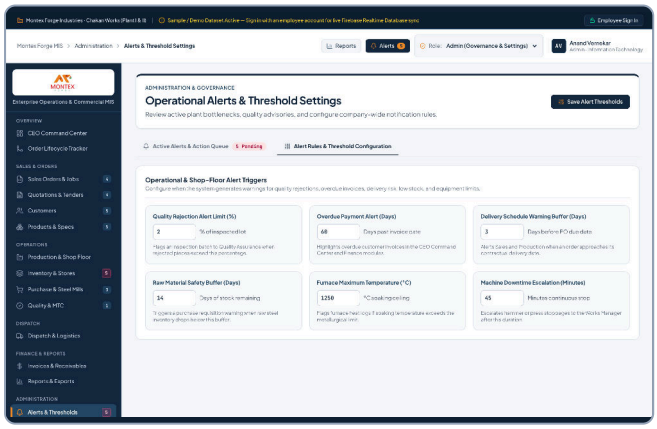
- Switch across all 8 report categories with live data previews and summary totals.
- Export multi-sheet branded Excel workbooks (.xlsx), CSV data files, or formatted PDF packs.

Real Application Capture (1440×900): Central Reports & Exports Center offering 8 structured MIS reports and an authoritative 19-Document Readiness Matrix. *[Synthetic Preview Data]*

Business Value: Standardizes management reporting while preventing incomplete draft documents from being mistaken for statutory certificates.

7.12B Operational Alerts Queue & Threshold Rules (/actionQueue & /alertThresholds)

VERIFIED



Page Purpose: Configure plant-wide warning thresholds and manage open operational exceptions across Stores, Quality, Production, and Finance.

Primary Users: Super Admin (Threshold Config), All Department Heads (Alert Acknowledgement)

KEY INFORMATION SHOWN

- Configurable numeric thresholds: Max Furnace Temp (°C), Max Rejection Rate (%), Overdue Payment Days, Machine Downtime (Min), and Power (kWh/MT).
- Active Alerts & Action Queue tab (Screenshot 27) with severity filters and automated quantity-invariant advisories.

ACTIONS AVAILABLE

- Adjust and save company alert thresholds directly to Cloud Firestore (/alertThresholds).
- Acknowledge open operational alerts or drill down to the affected order/module.

Real Application Capture (1440×900): Company-wide Alert Threshold Configuration panel alongside the active Operational Exception Queue. *[Synthetic Preview Data]*

Business Value: Gives management control over plant tolerance limits without requiring code changes.

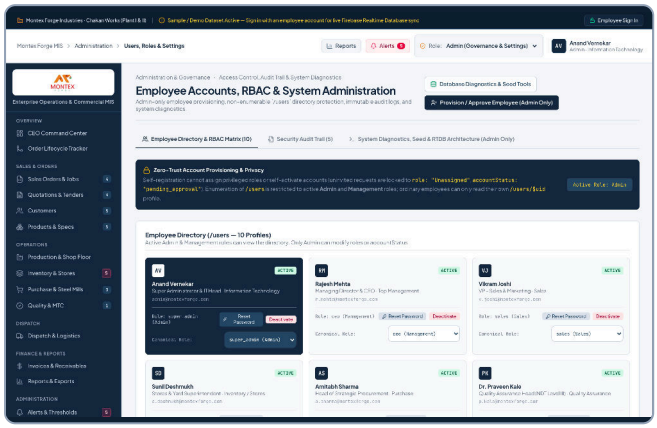
SECTION 7 — Product Tour: Users, 8-Role RBAC, Audit Trail & Architecture

Zero-trust employee account provisioning, 8 canonical roles, immutable security audit logs, and database diagnostics.

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7.13A Employee Directory & 8-Role RBAC Matrix (/users)

VERIFIED



Page Purpose: Provision employee accounts securely, assign canonical roles, deactivate departing staff, and enforce module-level permissions.

Primary Users: Super Administrator (Full Control), CEO / Management (Read-Only Directory)

KEY INFORMATION SHOWN

- Employee profile cards showing Name, Email, Department, Designation, Canonical Role, and Active/Deactivated status.
- 8 Canonical Roles: super_admin, ceo, sales, stores, purchase, quality, dispatch, and finance.
- Built-in Self-Lockout Guard and Last-Super-Admin Guard preventing accidental administrative lockout.

ACTIONS AVAILABLE

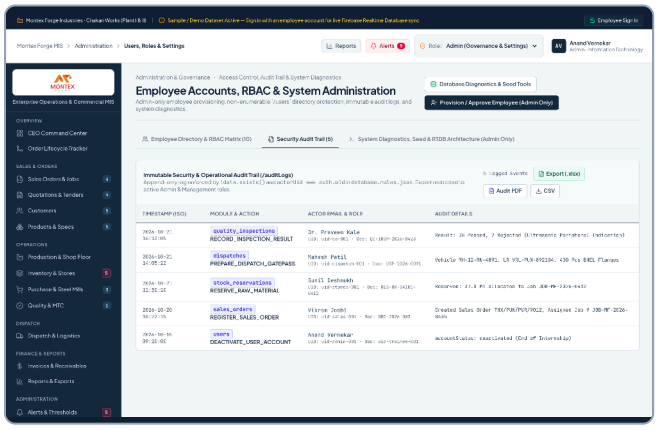
- Provision new employee accounts via server-side Firebase Admin SDK (/api/admin/employees).
- Update an employee's canonical role, toggle active/deactivated status, or trigger a safe password-reset flow.
- Export the Employee Directory, RBAC Matrix, and Security Audit Governance Pack to Excel (.xlsx).

Real Application Capture (1440×900): Admin governance console showing active employee profiles, canonical role assignment, password reset, and RBAC matrix. *[Synthetic Preview Data]*

Business Value: Ensures employees only view or modify data belonging to their department, eliminating unauthorized edits.

7.13B Immutable Security Audit Trail & Cloud Architecture (/auditLogs)

VERIFIED



Page Purpose: Maintain a tamper-resistant record of sensitive administrative and operational actions alongside live Firestore diagnostics.

Primary Users: Super Administrator, CEO, Internal / Statutory Auditors

KEY INFORMATION SHOWN

- Chronological audit entries: Audit ID, Timestamp, Actor Name/UID, Target Collection & Document ID, Action, and Summary.
- Strict secret redaction ensuring passwords, tokens, and private keys are never written to audit logs.
- System Diagnostics & Architecture tab (Screenshot 31) displaying Firestore collection health and security rules.

ACTIONS AVAILABLE

- Inspect and filter security audit logs or export them to CSV for compliance reviews.
- Open the Firebase Auth & Diagnostics Modal to verify live database connectivity and rules.

Real Application Capture (1440×900): Append-only Security Audit Trail recording actor UID, target collection, action type, timestamp, and outcome. *[Synthetic Preview Data]*

Business Value: Provides complete accountability for role changes, account deactivations, stock adjustments, and order updates.

SECTIONS 8 & 9 — Role-Based Access Security & Cloud Firestore Persistence

How the 8 canonical employee roles, security safeguards, and persistent operational registers protect company data.

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Section 8 — Canonical 8-Role Access & Permission Governance Matrix

VERIFIED

CANONICAL ROLE	RBAC GROUP	ACCESSIBLE MODULES (VIEW)	AUTHORIZED WRITE / CREATE / EDIT ACTIONS	EXPORT & ADMIN SCOPE
`super_admin`	**Admin**	All 16 Modules + Admin, Audit Trail & Diagnostics	Full write access across all collections + Employee provisioning, role assignment, account activation & alert thresholds	All PDF/XLSX/CSV exports + Admin Governance Pack
`ceo`	**Management**	CEO Dashboard, Order Journey, All Commercial, Plant & Finance Views, Read-Only Employee Directory	Executive read-only oversight + Alert acknowledgement (prevents accidental operational edits by top management)	All Executive, Sales, Plant & Finance PDF/XLSX exports
`sales`	**Sales**	Dashboard, Order Journey, Sales Orders, Quotations, Tenders, Customers, Products, Quality/Dispatch (View)	Create/update Quotations, Tenders, Customer POs (`/salesOrders`), Customer & Product Masters, Order stage updates	Quotations, Tenders, Orders & Customer exports
`purchase`	**Production**	Purchase & Steel Mills, Inventory & Stores, Suppliers, Order Journey	Issue Steel Mill Purchase Orders (`/purchaseOrders`), record Mill GRN receipts, update Supplier records	Procurement, Mill PO & Shortage Indent exports
`stores`	**Production**	Inventory & Stores, Purchase/GRN, Production Shop Floor, Order Journey	Record Raw Material Stock Movements (`/stockMovements`), Mill GRNs, Machine/Furnace/Maintenance/Power logs	Inventory, GRN Voucher & Shop-Floor exports
`quality`	**Production**	Quality & MTC, Dimensional/TPI, Production, Order Journey, Products	Record Lot Quality Inspections (`/qualityInspections`), Dimensional Reports (`/dimensionalReports`), TPI clearance status	Quality Lot, DIR & Draft MTC Summary exports
`dispatch`	**Production**	Dispatch & Logistics, Order Journey, Sales Orders (View), Quality (View)	Record Outward Dispatches & LR consignments (`/dispatches`) and update Order dispatch stage	Delivery Challan PDF & Dispatch LR Register exports
`finance`	**Finance**	Finance & Receivables, 8-Head Job Costing, Tally Sync, Sales Orders, Purchase Orders, Reports	Record Customer Payment Receipts (`/outstandingAgeing`), update Invoice status & Job Costing sheets	Receivables, Payment Voucher, Costing & Invoice exports

5 Verified Security & Governance Safeguards

- **1. Active Account Enforcement:** Both frontend guards and `firestore.rules` verify `accountStatus === 'active'`. Deactivated employees are immediately blocked from reads, writes, and exports.
- **2. Server-Side Employee Provisioning:** Only an active `super_admin` can create accounts or assign roles via `/api/admin/employees`. Self-registration cannot self-assign privileged roles.
- **3. Self-Lockout & Last-Admin Protection:** Built-in guards prevent a Super Admin from deactivating their own account or demoting the last active Super Admin.
- **4. Non-Enumerable User Directory:** Collection-wide listing of `/users` is restricted to active `super_admin` and `ceo` roles; other staff can only read their own `/users/{uid}` profile.
- **5. Secret-Redacted Immutable Audit Logs:** Sensitive operations write append-only records to `/auditLogs` with automatic redaction of passwords, tokens, and private keys.

Section 9 — Cloud Firestore Persistence & 8 Operational Registers

VERIFIED

In Phases 4 and 5, the application's data foundation was upgraded to **Cloud Firestore** with a controlled cutover engine (`MontexCutoverEngine`) and bidirectional schema mappers:

- **All 8 Operational Registers Are Now Persistent:** `/quotations`, `/tenders`, `/purchaseOrders`, `/machineLogs`, `/furnaceLogs`, `/maintenanceTickets`, `/powerLogs`, and `/dimensionalReports` persist directly to Cloud Firestore alongside the 11 core collections.
- **Atomic Multi-Document Transactions:** Stock movements, Mill GRN receipts, and outward dispatches use atomic Firestore transactions so inventory balances and PO/order balances never go out of sync.
- **Strict Demo vs. Production Isolation:** Unauthenticated preview sessions use labeled synthetic data; authenticated production sessions read strictly from Cloud Firestore and display clean empty states when a collection is empty (never mixing demo data into live records).

SECTIONS 10 & 11 — Business Benefits for Montex & Honest Current Limitations

Expected operational value by department and transparent disclosure of items requiring live plant confirmation.

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Section 10 — Expected Business & Operational Benefits by Department

Presented as expected operational improvements (subject to plant adoption)

1. Management & CEO Oversight

- Instant single-screen visibility into daily dispatches, order backlog, critical steel shortages, and 90+ day receivables.
- KPI Reconciliation panel verifies whether summary numbers match underlying order, stock, and invoice records.

2. Sales, Quotations & Customer Service

- Faster RFQ turnaround using standardized forged product weights and per-kg rate calculators.
- Immediate customer PO stage answers via the 7-Stage Order Lifecycle Tracker without phone follow-ups.

3. Purchase & Raw Material Stores

- Clear separation of Physical Stock vs. Job-Reserved Stock prevents unexpected billet shortages on forging hammers.
- Closed-loop Mill PO to GRN receipt updates yard stock and heat-number ledgers in one atomic step.

4. Production, Furnace, Maintenance & Power

- Digital shift logs, furnace fuel/MT tracking, and daily kWh/MT logs expose energy waste and press downtime root causes.
- Configurable alert thresholds flag furnace over-temperature and specific power spikes immediately.

5. Quality Assurance, NDT & TPI

- Enforced quantity invariants and structured Dimensional Inspection Reports (DIR) keep TPI and MTC records audit-ready.
- End-to-end linkage from Steel Mill Heat Number to final inspection lot reduces audit preparation effort.

6. Dispatch, Finance & Cost Control

- Outward dispatch logs automatically update pending PO piece balances and generate branded Delivery Challans.
- 8-head job costing and 0–90+ day ageing schedules protect order margins and prioritize overdue collections.

Section 11 — Honest Current Limitations & Items Requiring Live Confirmation

REQUIRES VERIFICATION

To ensure complete transparency for Montex management, the following items are not yet active in a live production environment and require credentials, company master data, or Phase-2 enhancements before full plant cutover:

AREA / COMPONENT	CURRENT STATUS	HONEST LIMITATION & WHAT IS REQUIRED BEFORE PRODUCTION CUTOVER
1. Server Admin SDK Credentials	REQUIRES VERIFICATION	The backend employee provisioning routes (`/api/admin/*`) and system init script are implemented and tested against mocks, but require live Firebase Admin service-account credentials (`FIREBASE_SERVICE_ACCOUNT_JSON`) in the production hosting environment.
2. Cloud Firestore Rules Deployment	REQUIRES VERIFICATION	`firestore.rules` are written and reviewed locally for all 19 collections and 8 roles, but must be deployed to the live Firebase project and verified with authenticated employee accounts before production traffic is switched.
3. Statutory Tax Invoice & e-Way Bill	PARTIALLY IMPLEMENTED	The system generates a clearly labeled **Draft Commercial Billing Summary PDF**, but does not yet integrate with the NIC GST e-Invoice portal (IRN/QR code generation) or automated e-Way Bill APIs.
4. Final EN 10204 3.1 MTC Lab Values	PARTIALLY IMPLEMENTED	The system generates a **Draft MTC Data Summary PDF** (linking heat number, UT/MPI, hardness, and DIR), without fabricating chemical composition (%C, %Mn, %Cr, %Mo) or tensile/impact lab test numbers, which require spectrometer/UTM data entry fields in Stage A.
5. Tally Prime Integration Mode	PARTIALLY IMPLEMENTED	Currently supports **CSV/XLSX file upload reconciliation** and CSV export. Direct two-way Tally Prime XML/HTTP bridge synchronization is not yet connected and requires confirmation of Montex's Tally ledger names and voucher types.
6. External Email / WhatsApp Alerts	PLANNED	Operational alerts and threshold breaches are displayed and acknowledged inside the web application's **Alerts & Action Queue**. Automated external email/WhatsApp push notifications and scheduled daily report emails are planned for Stage B.
7. Shop-Floor IoT & Sensor Input	PLANNED	Furnace temperatures, shift production counts, and daily kWh readings are currently entered via validated web forms by supervisors. Direct PLC, pyrometer, and smart energy-meter telemetry is a Stage C roadmap item.

SECTIONS 12, 13 & 14 — Future Digital Transformation Roadmap (Stages A, B & C)

Phased, practical recommendations to take Montex Forge from initial live cutover to advanced industrial intelligence.

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Section 12 — Stage A: Immediate Production Readiness (Next 2–4 Weeks)

REQUIRES VERIFICATION

RECOMMENDED FEATURE	TARGET USERS	BUSINESS BENEFIT	COMPLEXITY	KEY DEPENDENCIES
A1. Live Cloud Rules & Admin Cutover	IT Admin, All Staff	Activates real multi-user authentication, server-side employee provisioning, and live Cloud Firestore security rules.	**Low**	Production Firebase Admin credentials & rules deploy.
A2. Master Data & Opening Stock Onboarding	Sales, Stores, Purchase, Finance	Populates real Montex customers, forged products, steel suppliers, billet yard stock by heat, and open POs.	**Low**	Verified Excel/CSV master lists from Montex teams.
A3. Full EN 10204 3.1 Lab Certificate Fields	Quality / Metallurgy Lab	Upgrades Draft MTC Summary to complete EN 10204 3.1 MTC with ladle/check chemical % and tensile/yield/impact values.	**Medium**	Approved Montex MTC format & grade spec table.
A4. Multi-User Departmental UAT Sign-Off	All 8 Department Leads	Validates real-world workflows, print layouts, and role boundaries across 1–2 live customer orders before plant-wide rollout.	**Low**	Nomination of 1 champion user per department.

Section 13 — Stage B: High-Impact Operational Automation (1–3 Months)

PLANNED

RECOMMENDED FEATURE	TARGET USERS	BUSINESS BENEFIT	COMPLEXITY	KEY DEPENDENCIES
B1. Automated Morning CEO & Plant Email/WhatsApp Digest	CEO, MD, Plant Head	Delivers an automated 8:30 AM summary of yesterday's dispatches, shift tonnage, furnace fuel, and critical alerts.	**Medium**	SMTP / WhatsApp Business API credentials.
B2. Barcode / QR Billet & Heat Tagging	Stores, Cutting, Forging Shop	Prints QR heat tags at GRN receipt so operators scan billet bundles on the shop floor, eliminating heat-number typos.	**Medium**	Industrial label printer & mobile/tablet camera scan.
B3. Customer / TPI Document Portal View	Sales, TPI Inspectors, Key Clients	Provides a secure, token-scoped read-only link for TPI agencies to review DIRs, MTCs, and lot status for their PO.	**Medium**	Management approval of external TPI sharing rules.
B4. Two-Way Tally Prime XML / Voucher Bridge	Finance & Accounts	Automates sales invoice, purchase GRN, and receipt voucher push/pull with Tally Prime without manual CSV uploads.	**Medium**	Tally Prime ledger mapping & local connector agent.

Section 14 — Stage C: Advanced Industrial Intelligence (3–6+ Months)

PLANNED

RECOMMENDED FEATURE	TARGET USERS	BUSINESS BENEFIT	COMPLEXITY	KEY DEPENDENCIES
C1. Furnace Pyrometer & Smart Energy Meter IoT Sync	Production, Electrical, Quality	Captures live furnace soaking curves (°C), PNG gas flow, and HT panel kWh/kVA automatically without manual form entry.	**High**	RS-485 / Modbus gateway on furnaces & HT meters.
C2. Die Life & Tooling Shop Management	Tool Room, Production, Costing	Tracks forging die blow counts, re-sinking cycles, and die cost amortization per forged piece across hammers/presses.	**Medium**	Tool-room die numbering catalog & life norms.
C3. Predictive Delivery Risk & Capacity Scheduler	Production Planning, Sales, CEO	Forecasts PO delivery dates based on hammer/ring-mill queue load, furnace availability, and incoming steel mill POs.	**High**	3+ months of historical shift & cycle-time data in MIS.
C4. AI Executive Briefing & Cost Anomaly Assistant	CEO, Finance Head, Plant Head	Highlights unusual job margin drops, supplier price hikes, or recurring machine breakdown patterns in plain language.	**Medium**	Verified historical Firestore dataset & Gemini API policy.

SECTIONS 15, 16 & 17 — Recommended KPIs, Rollout Roadmap & Information Needed

Core metrics Montex leadership should review, the 6-step go-live sequence, and the onboarding checklist.

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Section 15 — Recommended Management KPIs Enabled by the MIS

DOMAIN	KPI NAME	FORMULA / OPERATIONAL MEANING	OWNER & CADENCE	MANAGEMENT DECISION ENABLED
Commercial	Quotation Win Rate & Price/Kg	Won Quote Value / Total Quoted Value; Avg Realized Rs./Kg	Sales Head · Weekly	Adjust bidding margins and prioritize high-conversion customers/grades.
Order Delivery	On-Time Delivery (OTD %)	Orders Dispatched <= Contractual PO Date / Total Dispatched Orders	CEO / Planning · Weekly	Reallocate hammer shifts or expedite TPI witnessing for at-risk POs.
Stores & Purchase	Net Free Billet Coverage (MT)	Physical Stock MT minus Reserved Job Stock MT vs. Reorder Level	Stores / Purchase · Daily	Trigger Steel Mill POs before alloy steel deficits halt production.
Production	Press Utilization & Fuel/MT	Run Hours / Available Shift Hours; Furnace PNG SCM or LDO Ltr / MT	Plant Head · Daily	Optimize furnace batch loading and reduce die-changeover downtime.
Energy	Specific Energy (kWh / MT)	Daily Plant HT Electricity (kWh) / Daily Forged Output (MT)	Electrical · Daily	Control energy cost per ton and maintain Power Factor > 0.98.
Quality	First-Pass Yield & COPQ (%)	QC Passed Pcs / Total Inspected Pcs; Scrap & Rework Cost	QA Head · Weekly	Correct die wear, overheating, or dimensional drift on specific jobs.
Finance	60+ / 90+ Day Ageing & Job Margin	Receivables > 60 Days (Rs.); (Invoiced Revenue - 8-Head Cost) / Revenue	Finance / CEO · Weekly	Enforce credit holds on overdue accounts and re-price low-margin jobs.

Section 16 — 6-Step Practical Rollout Roadmap

- **Step 1 — Master Data & Rules Setup (Week 1):** Deploy `firestore.rules`, configure live Firebase Admin credentials, and import verified Customers, Products, and Suppliers.
- **Step 2 — Provision Employee Accounts (Week 1):** Super Admin provisions named employee logins across all 8 roles via the Admin Console and verifies role access.
- **Step 3 — Pilot Sales, Stores & Purchase (Week 2):** Go live with Quotations, Customer PO registration, Billet Yard opening balances, and Steel Mill PO/GRN entry.
- **Step 4 — Enable Shop-Floor, Quality & Dispatch (Week 3):** Train shift supervisors, QA engineers, and dispatch officers to log daily shifts, furnace batches, DIRs, and vehicle LRs.
- **Step 5 — Activate Finance, Costing & CEO Reviews (Week 4):** Load receivables ageing, run weekly CEO Command Center reviews, and export official PDF/Excel packs.
- **Step 6 — Phase-2 Automation (Month 2+):** Add full lab MTC fields, scheduled email/WhatsApp digests, and Tally XML sync.

Section 17 — Information Requested from Montex

To transition the system from synthetic preview data to live plant operation, we request the following inputs from Montex management:

- **1. Employee Access List:** Names, official email addresses, designations, and assigned role (from the 8 canonical roles) for initial users.
- **2. Customer & Supplier Masters:** Current Excel list of active customers (with credit days/limits) and approved steel mills.
- **3. Forged Product Catalog:** Standard item codes, grades, finished weights (kg), gross input billet weights (kg), and HSN codes.
- **4. Opening Billet Stock & Open POs:** Current yard stock by grade, section, and heat number, plus active customer & steel mill POs.
- **5. Sample Plant Documents:** 1 sample each of Montex's current Quotation, MTC (EN 10204 3.1), Dimensional Report, Delivery Challan, and Tax Invoice.
- **6. Alert Threshold Sign-Off:** Confirmation of target limits for furnace temp (°C), rejection %, overdue days, downtime, and kWh/MT.

SECTIONS 18 & 19 — Executive Glossary & Closing Transformation Summary

Plain-language reference of technical and forging MIS terms and recommended next steps for Montex leadership.

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Section 18 — Glossary of Key Technical & Forging Operations Terms

TERM / ACRONYM	SIMPLE BUSINESS & PLANT MEANING
MIS (Management Information System)	A centralized digital platform that collects operational and financial data from every department and turns it into live dashboards, alerts, and management reports.
RBAC (Role-Based Access Control)	A security mechanism where each employee's login role (e.g., Sales, Stores, Quality, Finance, CEO) strictly determines which screens they can view, edit, or export.
Cloud Firestore	Google Cloud's enterprise document database used by the Montex MIS to store all 19 operational collections persistently with real-time synchronization and security rules.
Mill Heat Number	The unique melt/cast batch identifier stamped by the primary steel mill (e.g., JSW, SAIL, Kalyani) on raw steel billets, providing chemical traceability through forging to final MTC.
GRN (Goods Receipt Note)	The official inward stores record confirming that a specific tonnage and heat number of raw steel billets has been received and inspected against a Steel Mill Purchase Order.
MTC (EN 10204 3.1 Certificate)	A statutory Metallic Material Test Certificate verifying that forged components comply with customer PO chemical, mechanical, heat-treatment, and non-destructive testing specs.
DIR (Dimensional Inspection Report)	Quality Assurance record comparing actual measured dimensions (Outer Diameter, Bore, Thickness, PCD) of sampled forgings against the approved engineering drawing tolerances.
TPI (Third-Party Inspection)	Independent inspection agency (such as Lloyds, Bureau Veritas, TUV, or EIL) appointed by the customer to witness testing and issue a dispatch clearance certificate.
Receivables Ageing Bucket	Grouping of unpaid customer invoices by how many days they have been outstanding (0–30, 31–60, 61–90, and 90+ days) to prioritize payment follow-up and credit control.
Immutable Audit Log	A tamper-resistant security ledger (`/auditLogs`) that permanently records who performed a sensitive action, what record was changed, and when it occurred.

SECTION 19 — CLOSING EXECUTIVE SUMMARY

Ready for Controlled Plant Onboarding & Executive Review

The **MONtex Forge Industries MIS Dashboard** establishes a complete, role-secured digital backbone across Sales, Tendering, Procurement, Billet Stores, Forging Shop Floor, Heat-Treatment Furnaces, Maintenance, Power Monitoring, Quality/MTC, Dispatch, and Finance. With **16 interactive modules**, **19 Cloud Firestore collections** (including all 8 persistent operational registers), and **31 passing automated verification suites**, the platform is ready to transition from synthetic preview data to a controlled pilot with real Montex plant data.

1. Review & Walkthrough
Conduct a 45-minute live walkthrough of the 16 modules with the MD, CEO, and department heads.

2. Share Onboarding Inputs
Provide the employee email/role list, master data sheets, and sample MTC/Invoice formats (Section 17).

3. Launch 2-Week Pilot
Deploy live Cloud Firestore rules and track initial live customer orders end-to-end across all 8 roles.

One Platform. Connected Operations. Better Decisions.

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